Tensor Network Simulation of the Heisenberg Model on Heavy-Hex Lattices

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July 10, 2026

Abstract

We extend CppSim, a high-performance C++/HIP tensor network simulator, to the heavy-hex lattice geometry used in superconducting quantum processors. The heavy-hex graph — a honeycomb of degree-2 and degree-3 sites with a natural 3-color gate schedule — is implemented as a drop-in grid class with no changes to the gate application or belief propagation (BP) kernels. We simulate the isotropic Heisenberg model $H = J\sum_{\langle ij\rangle}(\mathbf{X}_i\mathbf{X}_j + \mathbf{Y}_i\mathbf{Y}_j + \mathbf{Z}_i\mathbf{Z}_j)$ in the Heisenberg picture using Pauli transfer matrices, and study the autocorrelation $C(t) = \langle Z_c(t)Z_c(0)\rangle$ and operator lightcone on the `heavy_hex_3x3` graph (68 sites, 76 bonds). We demonstrate chi convergence at $\chi = 200$ ($|\text{error}| < 10^{-4}$, reference $\chi = 430$) on a 32 GB GPU, and show a $\sim$10$\times$ speedup over a Julia/AMDGPU.jl reference at saturated bond dimension.

1 Introduction

Heavy-hex is the lattice topology used in superconducting quantum processors [1]. Each qubit site has degree 2 or 3, and the graph admits a natural 3-coloring of its bonds, enabling parallel gate application within each color class.

Classical tensor network (TN) simulation of quantum circuits on heavy-hex graphs serves two purposes: (i) validation of quantum computations for small system sizes where exact simulation is feasible [2, 3, 4, 5], and (ii) computation of noise-free reference values for quantum error mitigation protocols such as the Multi-Product Formula (MPF) [6].

In this paper we show that CppSim [7], originally developed for rectangular 2D Ising dynamics, extends naturally to heavy-hex topology with minimal code changes. The hot-path kernels — gate application, BP, workspace management — are topology-agnostic and require no modification. Only the grid construction and Hamiltonian gate matrix need to be defined for the new geometry.

2 Heavy-Hex Lattice and Hamiltonian

We describe the graph topology, its implementation in CppSim, and the Hamiltonian gate matrix used for time evolution.

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2.1 Graph structure

The `heavy_hex`$_{n_x} \times n_y$ graph has:

- $(n_x + 1)(n_y + 1)$ *junction* sites (degree 3)
- remaining *arm* sites (degree 2) connecting adjacent junctions
- $2n_xn_y + n_x + n_y$ bonds total

For $n_x = n_y = 3$: 16 junction sites, 52 arm sites, 76 bonds. The graph is bipartite and 3-edge-colorable — bonds partition into 3 color classes of sizes 25, 25, 26, corresponding to the natural gate layers of the hardware gate schedule.

Site tensors have shapes $[\chi^2, D]$ (arm) and $[\chi^3, D]$ (junction), where $D = 4$ is the Pauli basis dimension $\{I, X, Y, Z\}$. The memory cost is dominated by junction sites: $16 \times \chi^3 \times D \times 4\,\mathrm{B}$ (float32).

2.2 Implementation

The `FlatHeavyHexGrid` class inherits from `Grid` and populates the same `Site`/`Bond`/`Gate` data structures used by the rectangular grid. The color class assignment follows the hardware gate schedule (JSON layers 0, 1, 2). No changes are required in `gate.hpp`, `measures.hpp`, or `workspace.hpp` beyond parametrizing the workspace by `max_degree=3` instead of 4.

2.3 Hamiltonian and gate matrix

We simulate the isotropic Heisenberg (XXX) model:

$$H = J \sum_{\langle ij \rangle} \left(X_i X_j + Y_i Y_j + Z_i Z_j \right) \tag{1}$$

with uniform coupling $J = 1$ and time step $\delta t = 0.1$.

In the Heisenberg picture with Pauli basis, the two-site gate $U = e^{-i\theta(X\otimes X + Y\otimes Y + Z\otimes Z)}$ ($\theta = J\delta t$) is represented by a 16×16 real Pauli transfer matrix (PTM) with block structure:

$$\{II, XX, YY, ZZ\} \to \text{invariant (diagonal} = 1) \tag{2}$$
$$\{IX, XI, YZ, ZY\} \to 4 \times 4 \text{ mixing block} \tag{3}$$
$$\{IY, YI, XZ, ZX\} \to 4 \times 4 \text{ mixing block} \tag{4}$$
$$\{IZ, ZI, XY, YX\} \to 4 \times 4 \text{ mixing block} \tag{5}$$

with elements $\alpha = \cos^2(2\theta)$, $\beta = \sin^2(2\theta)$, $\gamma = \sin(2\theta)\cos(2\theta)$. This was verified against a Julia reference implementation.

3 Autocorrelation and Chi Convergence

We initialize the Heisenberg-picture operator as Z at a single center site (site 13, a degree-3 junction at the graph midpoint) and I elsewhere. The autocorrelation $C(t) = \langle Z_c(t) Z_c(0) \rangle$ is computed via the Bethe approximation [2, 8] after each Trotter step. We study convergence with respect to the bond dimension χ by running nine simulations with $\chi \in \{50, 100, 150, 200, 250, 300, 350, 400, 430\}$ over 30 Trotter steps at cutoff 10^{-4}.

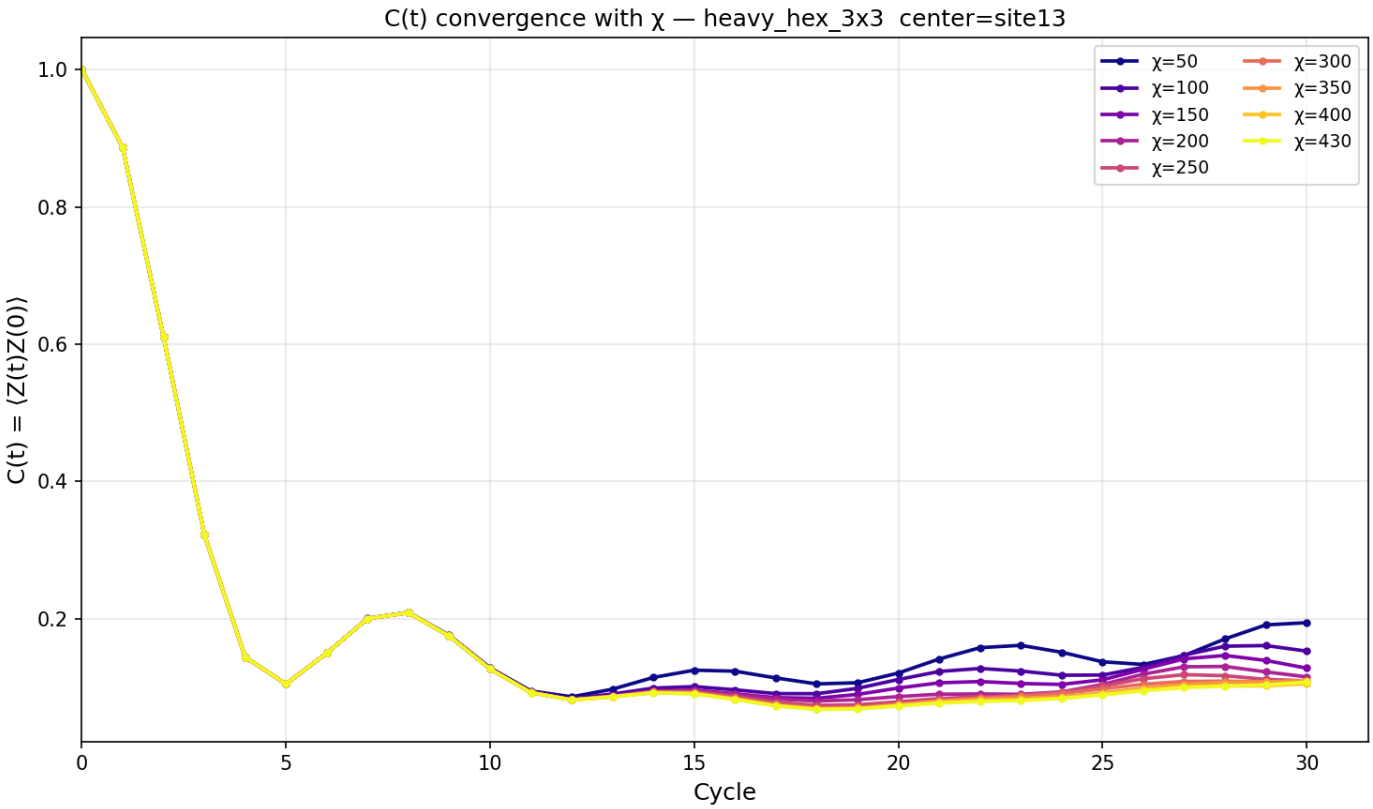


Figure 1: Autocorrelation $C(t) = \langle Z_c(t) Z_c(0) \rangle$ for $\chi = 50, 100, 150, 200, 250, 300, 350, 400, 430$ over 30 Trotter steps. The oscillatory behaviour is a genuine quantum recurrence of the XXX model on the finite heavy-hex graph. Curves for $\chi \geq 150$ are visually indistinguishable, indicating convergence.

3.1 Convergence with χ

Figure 1 shows $C(t)$ for $\chi = 50, 100, 150, 200, 250, 300, 350, 400, 430$ over 30 Trotter steps. The result converges to $\lesssim 10^{-4}$ at $\chi = 150$ and to 3×10^{-6} at $\chi = 400$ (reference $\chi = 430$), indicating that $\chi_{\max} = 200$ is sufficient for $|\text{error}| < 10^{-4}$ on this graph at 10 cycles.

The non-monotonic behaviour of $C(t)$ — a decrease followed by partial recovery — is a genuine quantum recurrence of the XXX model on the degree-3 lattice, confirmed by its independence from χ (Fig. 2).

3.2 Operator lightcone

Figure 3 shows the non-identity Pauli weight $n(v, t)$ at each site for 20 cycles. The operator spreads radially outward from the center junction, following the 3-fold symmetry of the hexagonal lattice. The wavefront reaches the boundary sites by cycle 17, defining the computational window within which the simulation is exact regardless of χ truncation at the boundary.

4 Performance

We report wall-clock times on a 32 GB GPU using Householder QR and float32 arithmetic. All timings are per Trotter cycle at saturated bond dimension.

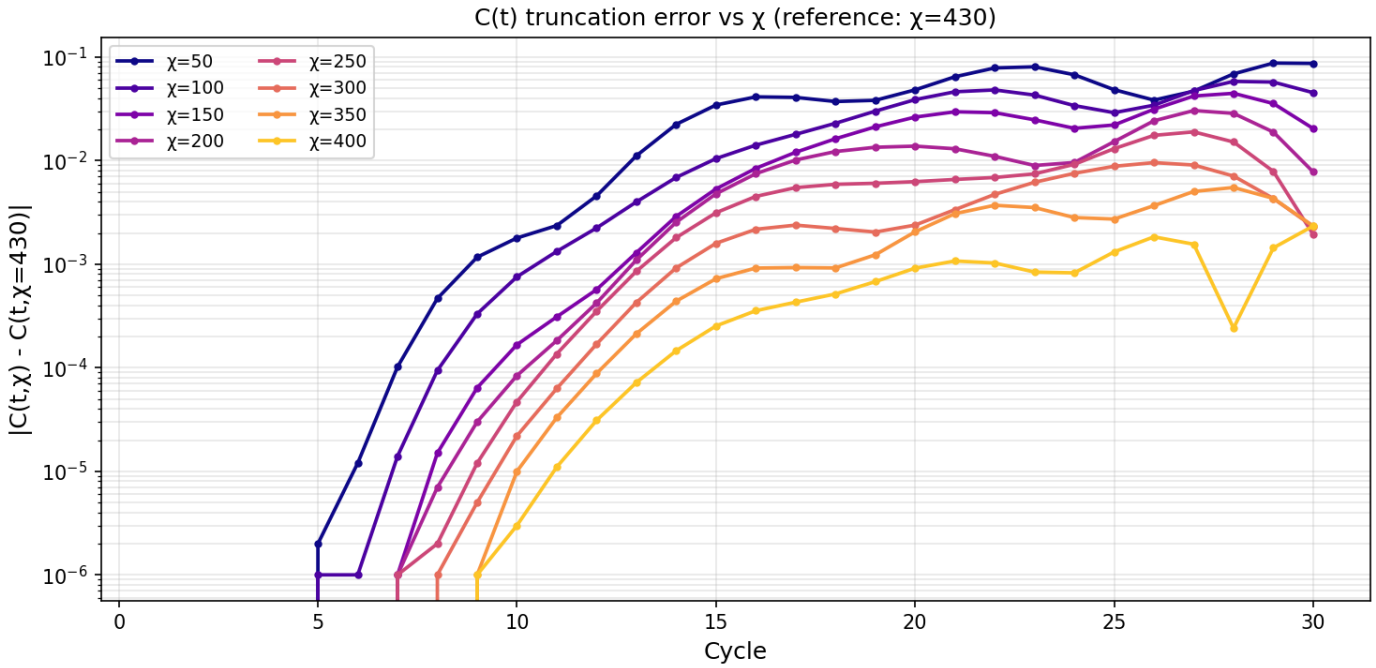


Figure 2: C(t) truncation error $|C(t,\chi) - C(t,\chi = 430)|$ vs cycle for each χ value. Errors onset when the bond dimension first saturates at the cap (visible as the left edge of each curve), then grow as more bonds are affected. At cycle 10: $\chi = 400$ gives error 3×10^{-6}, $\chi = 150$ gives 1.7×10^{-4}.

4.1 Chi scaling

Figures 4 and 5 show bond dimension growth and compute cost. All runs saturate at their cap by cycle 7-9. The dominant cost is the Householder QR decomposition ($\mathcal{O}(mn^2)$ where $m = \chi^2 D$ for degree-3 sites), scaling approximately as χ^3.

4.2 CppSim vs Julia/AMDGPU.jl

Table 1 compares wall-clock times at $\chi = 430$, cutoff=10^{-4}. Julia times report autocorrelation evolution only (excluding overlap computation). At saturated bond dimension (cycles 8–10), CppSim achieves a consistent $\sim$10$\times$ speedup.

At saturated bond dimension ($\chi = 430$, cycles 8–10), CppSim achieves a $\sim$10$\times$ speedup over the Julia/AMDGPU.jl reference implementation on the same GPU. Both use Householder QR and float32 arithmetic. The speedup arises from: (i) zero GPU memory allocation during simulation (pre-allocated workspace), (ii) direct HIP/rocBLAS calls without framework dispatch overhead, and (iii) on-device `pseudo_sqrt` kernel eliminating CPU–GPU transfers.

5 Conclusion

We have demonstrated that CppSim, a C++/HIP tensor network simulator, extends to the heavy-hex lattice with minimal code changes. The topology-agnostic gate and BP kernels accept any `Grid`-derived class; the heavy-hex extension requires only a new site/bond/color construction and a 16×16 Pauli transfer matrix for the XXX Hamiltonian.

Table 1: Wall-clock time per cycle. GPU runs: χ capped at 430, float32, Householder QR, cutoff=10^{-4}. CPU: float64, no χ cap (grows freely to χ_{CPU}). Julia: GPU, noiseless evolution time only. '—': not measured.

Cycle	$\chi_{\max}$	CPU (s)	χ_{CPU}	Julia (s)	CppSim (s)	Speedup
2	14	9	14	29	0.1	290×
4	60	15	58	29	0.5	58×
6	165	201	202	38	1.9	20×
7	278	978	364	65	4.4	15×
8	430	4,393	637	132	12.3	11×
9	430	—	—	251	25.2	10×
10	430	—	—	418	46.9	9×

On a 32 GB GPU, the 3×3 heavy-hex graph (68 sites, 76 bonds) is simulated with bond dimension $\chi \leq 430$. The autocorrelation $C(t)$ converges at $\chi = 200$ ($|\text{error}| < 10^{-4}$ relative to $\chi = 430$) for 10 Trotter cycles. The non-monotonic behaviour of $C(t)$ — a quantum recurrence of the isotropic Heisenberg model on the finite graph — is confirmed independent of χ and is therefore a physical prediction of the simulation. We demonstrate a $\sim$10× speedup over a Julia/AMDGPU.jl reference [5] at saturated bond dimension.

The operator lightcone, computed as a byproduct with no additional cost, visualizes operator spreading on the hexagonal lattice and defines the computational window within which the simulation is exact.

Future work will extend to larger graphs on higher-memory hardware, and implement interleaved BP to match the online approach of [2] and eliminate BP trap effects.

References

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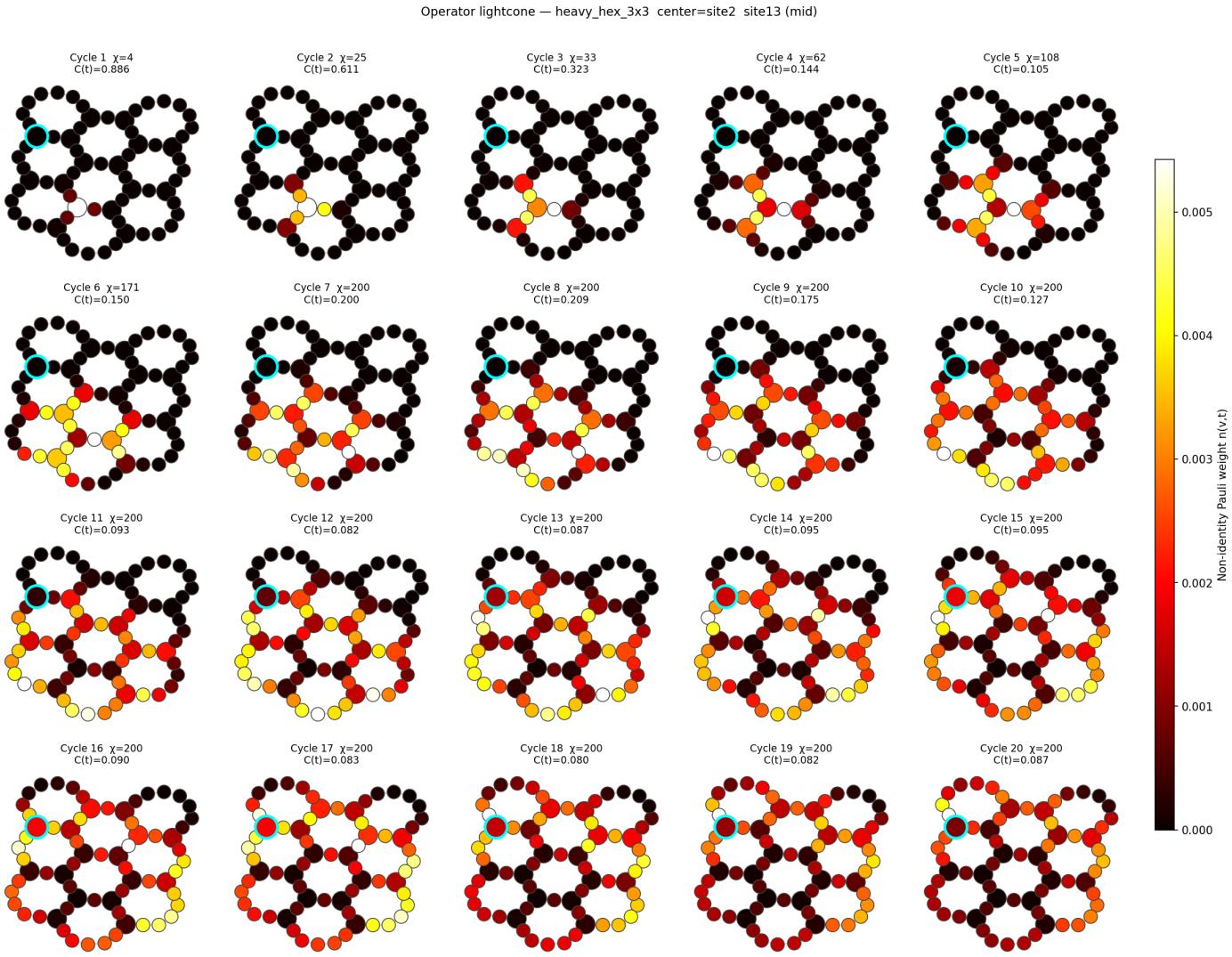


Figure 3: Non-identity Pauli weight $n(v,t)$ at each site of the heavy_hex_3x3 graph over 20 Trotter steps, initialised at site 13 (mid-graph degree-3 junction, marked cyan). The operator spreads radially outward with 3-fold hexagonal symmetry, reaching the graph boundary by cycle $\sim$17. Color scale is per-panel normalized.

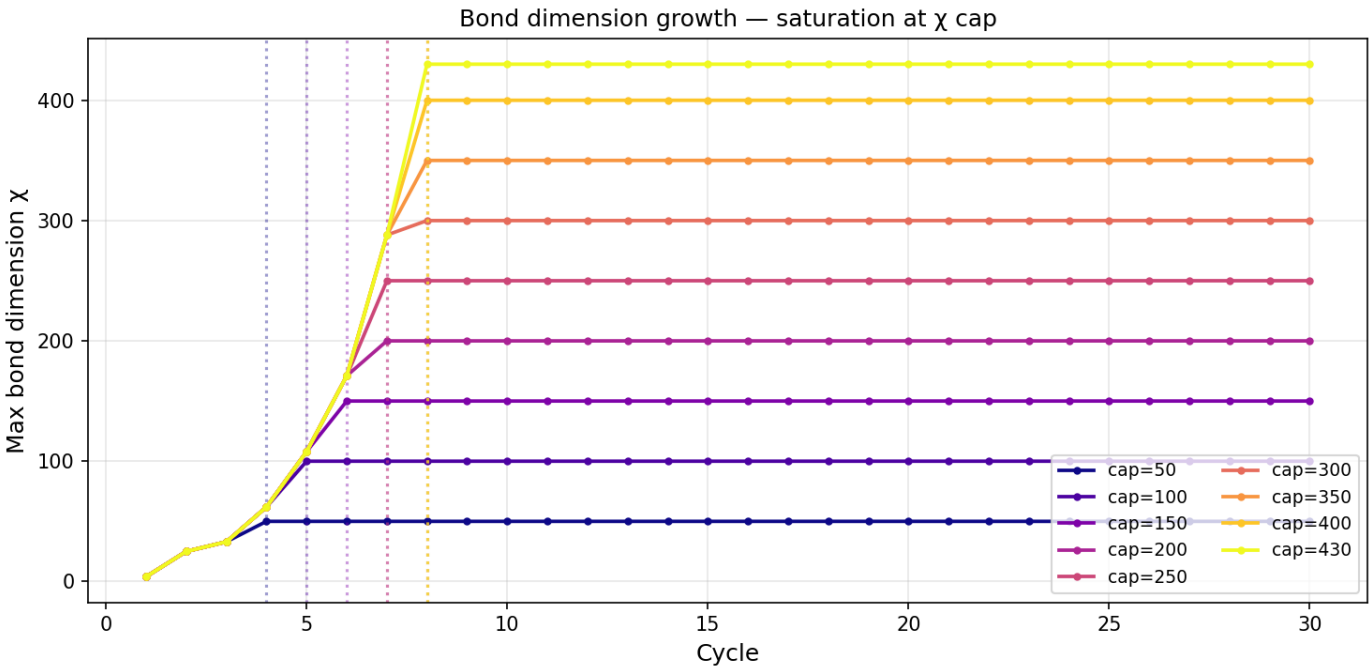


Figure 4: Maximum bond dimension vs cycle for each χ cap. All runs saturate at their cap by cycle 7-9. The dashed vertical lines mark the saturation cycle for each cap value.

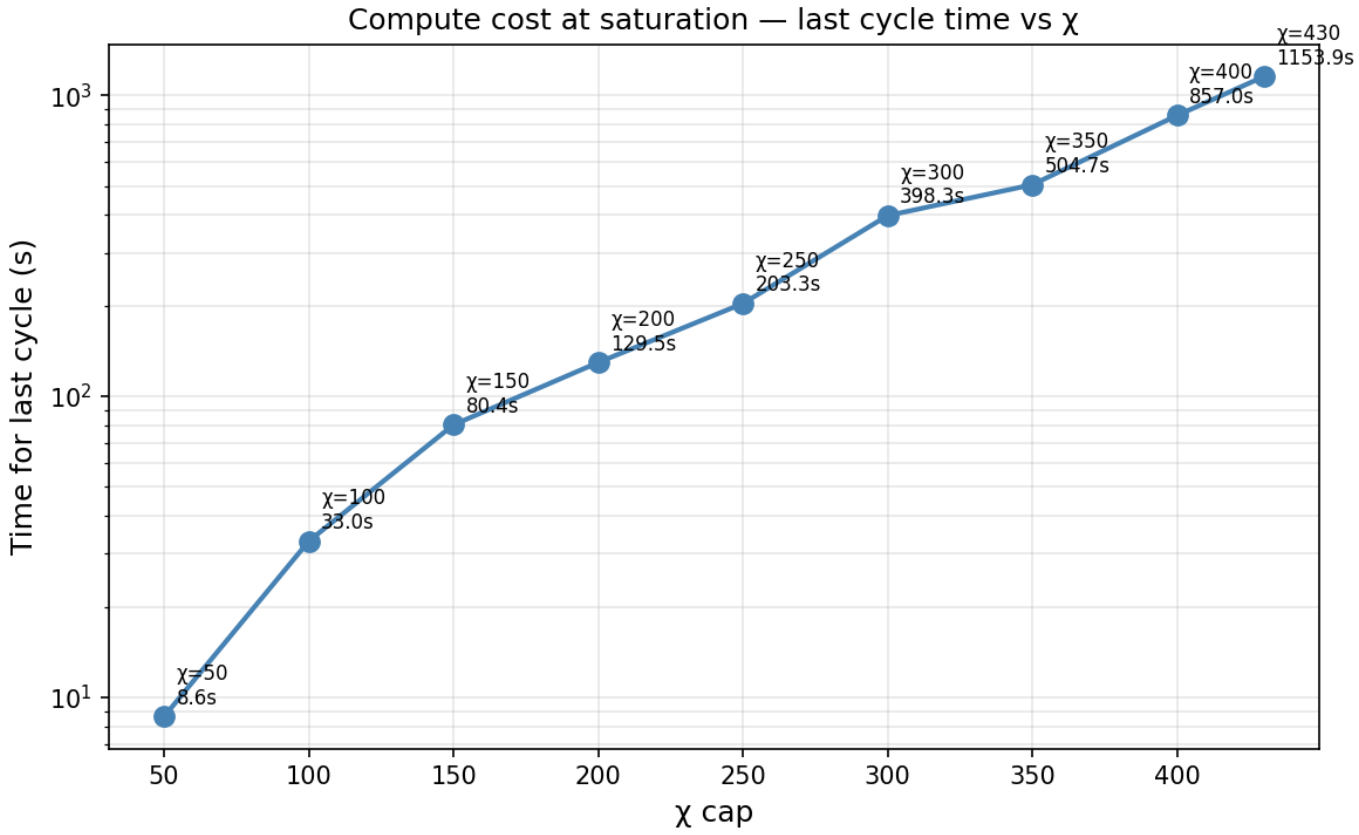


Figure 5: Wall-clock time of the last computed cycle vs χ cap, on a 32 GB GPU. Time scales approximately as χ^3 reflecting the Householder QR cost $\mathcal{O}(\chi^2 D \cdot (\chi D)^2)$ for degree-3 sites.